Deletion Without Absence: What a Modular Unlearner Answers Instead

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Abstract

Modular "exact unlearning" makes deletion cheap: train one small expert per data source, keep a selector that routes each query to an expert, and delete a source by dropping its expert. The guarantee is real but narrow — no surviving parameter was ever trained on the removed data — and it says nothing about the queries that used to reach the removed expert. Those queries still arrive, and the selector must send them somewhere. We audit where, on a 200-expert LoRA pool over TOFU with 20 of 200 sources deleted. Three findings. (i) The standard forget metric cannot see substitution: a "method" that deletes nothing and merely reroutes the deleted sources' queries to one arbitrary survivor scores at or above genuine deletion in six of seven destination choices at byte-identical utility, moving `forget_quality` by 0.53 on a choice unrelated to forgetting. (ii) The substitution is substantive: orphans are answered with a specific surviving author's distinctive facts at 0.24–0.30 against a 0.17 random-destination floor, and refusal essentially never happens (≤ 0.013 of 1600 answers). (iii) Every reassuring property of the pattern — orphan detectability, deletion locality, routing accuracy — is measured on queries that name the person being deleted, and collapses when the name is removed. A routed system's deletion claim is incomplete without reporting what it substitutes.

1 The queries survive the module

Machine unlearning asks a system to behave as though it had never seen some training data [Bourtoule et al., 2021]. Retraining a large language model is prohibitive, so a substantial line of work makes deletion cheap *architecturally*: partition the training data by source, train an isolated module per source, compose the modules at inference, and delete a source by dropping its module [Yu et al., 2022, Basu Roy Chowdhury et al., 2025, Zhao et al., 2024, Grimes et al., 2026, Schneider et al., 2026]. The guarantee is structural rather than statistical — afterwards, no surviving parameter was ever updated on the removed data — and it is narrower than it sounds. A modular system needs a *selector* to decide which module answers a query. Deletion removes a module; it does not remove the queries that were routed to it. Every one of those queries still arrives, and the selector must send it somewhere. We call such a query an *orphan*, and this paper asks one question: **what does the system say to an orphan?**

On a per-source expert pool over TOFU [Maini et al., 2024], it answers confidently with someone else's life. Asked whether a deleted author had won any awards, the system names an award — one belonging to a different, still-present author, with the name substituted and the rest of the sentence intact. It does this on roughly a third of orphaned queries and refuses on approximately none of them. The deletion was exact. The output is a fabricated biography attached to a real name, and the person whose facts were borrowed did not consent to that either.

Table 1: TOFU `forget_quality` across reroute destinations. Six of the seven arms that delete nothing score at or above genuine deletion, while `model_utility` is *identical to* 0.8009 *in all eight*.

Destination	s89	s137	s31	s97	**DELETE**	s33	s79	s88
expert affinity	.3044	.3382	.2840	.2267	—	.2663	.2193	.3970
`forget_quality`	**.8958**	.8002	.6288	.6288	**.5140**	.5140	.5140	.3615

Worse, the field's standard instrument scores this as success. TOFU's `forget_quality` is a two-sample test of *distributional indistinguishability* between the post-deletion model's forget-set behaviour and a reference model retrained without the data — and a stranger's answers are, in distribution, about as far from the deleted author's memorised answers as a retrained model's are. We exhibit an "unlearning method" that deletes nothing, only redirects, and scores at or above genuine deletion. The appendices carry the rest of the campaign, including four pre-registered hypotheses we refuted and eight measurement defects that produced plausible numbers before they were caught (App. I).

2 Setup: three deletions, three metrics

The pattern under audit. A frozen base model f_0; modules $\{\phi_1, \ldots, \phi_k\}$ with ϕ_i trained only on source i; a selector S mapping a query q to one module. The served answer is f_0 composed with $\phi_{S(q)}$, and deleting source i removes ϕ_i from the pool and from S's candidate set. This covers keyed adapter banks, sharded PEFT unlearning, retrieval-routed LoRA pools, per-user deletable proxies and block-listed memory modules (App. J). The selector is the part that has not been audited.

Three things "deleted" can mean. D1 (parametric): no parameter serving the query was trained on the deleted source — what modular architectures guarantee, by construction. **D2 (behavioural):** the system does not answer as though it still held the source — what a data subject takes "deleted" to mean. **D3 (indistinguishability):** an observer with query access cannot tell a deletion occurred. D1 implies neither, and this paper is about both gaps.

Metrics. TOFU supplies `forget_quality` and `model_utility`; neither is a statement about what the system substitutes. ORR is the fraction of orphaned queries answered substantively rather than refused, measured on generated text rather than on routing. CSAR is the fraction of orphan answers asserting at least one fact that is distinctive to the survivor the query was routed to, is *not* the deleted source's own fact, and is *not* in the frozen base's answer to the same question. RDR is the fraction of *retained* queries that change module as a side effect of a deletion nobody asked for. CSAR is deliberately not an n-gram metric: "Alice writes magical realism and was born in Lisbon" — Bob's genre and city, asserted about Alice — has low ROUGE-L against every gold answer in the corpus and is exactly the harm (App. B).

Pool, selectors, transforms. Llama-2-7B-chat frozen, $k = 200$ rank-32 LoRA experts [Hu et al., 2022], one per TOFU author; deletion is `forget10` (authors 180–199, 400 orphaned questions). Eight selectors in three families: lexical (`key_exact`, `key_tfidf`), feature-space (`centroid_sbert` over MiniLM embeddings [Reimers and Gurevych, 2019], `centroid_lm`), and behavioural (`ppl`, `activation_norm`, `attn_norm`, `logit_div`), the last of which *run* the candidate experts. Because a harm measured only on queries that name the deleted person is worth as much as a defense measured that way, every result is reported across a transform axis — `gold` (TOFU's question, which names the author in $\sim$90% of rows) versus `name_stripped` and three others — with the perturbation applied to *both* classes so no detector can learn "perturbed $\Rightarrow$ deleted" (App. A–C).

3 The forget metric is blind to substitution

We define an "unlearning method" that deletes nothing: all 400 orphaned queries are rerouted to one fixed surviving expert, while every expert — including the 20 supposedly forgotten ones — stays in the pool and in memory. Seven destination choices and one genuine-deletion baseline (experts

Table 2: Cross-source attribution over 400 orphan queries per router. "own-disclosure" is the rate at which the answer contains the *deleted* author's own facts — a different and worse failure, kept separate; "substantive" excludes rows whose matched facts are exhausted by name-forms of the routed survivor.

Condition	Router	CSAR	**substantive**	own-disclosure	refusal
`gold`	`centroid_sbert`	0.3325	**0.2400**	0.913	0.000
`gold`	`key_tfidf`	0.3650	**0.2950**	0.933	0.000
`name_stripped`	`centroid_sbert`	**0.4400**	—	0.273	≤0.010
`indirect`	`centroid_sbert`	0.3350	—	0.383	≤0.013
random destination	(no router)	0.2200	**0.1725**	0.953	0.003

dropped, orphans served by the frozen base) are scored at the same tier against the same held-fixed reference, so all eight cells are the same statistical test.

Table 1: six of seven arms that delete nothing score at or above genuine deletion, two land exactly on it, and `forget_quality` spans 0.3615–0.8958 — a swing of 0.53 — on a choice that holds fixed whether anything was forgotten, while `model_utility` does not move at all. The obvious explanation, that rerouting to a similar author makes the answers look right, is *refuted* (Spearman $\rho = +0.109$, $p = 0.82$, nominally opposite in sign to the hypothesis).

The arms score identical rows, so resolving the spread requires a *paired* interval; bootstrapping each arm's marginal re-adds the noise the arms hold in common and can never resolve any spread (App. I). Resampling row indices once per iteration and applying that draw to every arm gives a spread 95% CI of $[0.2245, 0.6975]$, $P(\text{spread} > 0.25) = 0.961$, and an at-or-above-deletion count of $6/7$ with 95% CI $[2, 7]$: even at the pessimistic end, two arms that delete nothing match or beat real deletion. Two limits travel with these numbers — the per-destination *ordering* does not reproduce across evaluation tiers, so no claim may name a winning destination, and a single `forget_quality` cell is worth only about ± 0.35 (App. D).

4 What the system says instead

Refusal never happens. At most 0.013 across 1600 answers: $ORR \approx 1.00$ at the level of what is *said*, not merely where the query is routed. The information needed to refuse is often present and then discarded — `key_exact` raises no-match on 100% of orphans, falls back to `candidates[0]`, and answers anyway.

It is not a naming glitch. The strongest objection to CSAR is that it counts the router emitting the survivor's *name* in place of the deleted author's. Splitting each cross-source row by whether its matched facts are exhausted by name-forms (Table 2), two thirds to four fifths of CSAR carries a real fact — a title, a place, an award — and it survives dropping the identity-question slice, where a wrong expert answers with a name by construction ($0.217/0.250$ substantive on the 300 non-identity queries). **We therefore quote the substantive figure,** 0.24**–**0.30 **against a** 0.17 **floor**, not the raw 0.33–0.37.

Anonymising the query does not reduce the harm; it changes its character. Strip the name and the system leaks *less* of the deleted author's own content ($0.913 \rightarrow 0.273$) while attributing *more* of a stranger's content to them ($0.3325 \rightarrow 0.4400$). Attribution does not require the router to be *right*, only confident enough to pick someone: whichever expert is activated asserts its own facts, and worse routing produces *more legible* attribution.

The floor matters. A uniformly random surviving expert — no router at all — already produces $CSAR = 0.2200$ (0.1725 substantive). The harm therefore **cannot be engineered to zero by improving the selector**; roughly two thirds of it is a property of substituting any expert at all. The quantity to report is the lift over that floor, not CSAR itself.

5 Every reassuring property is a lexical artifact

Granularity was the axis, and it does not survive anonymisation. The published finding that confidence-based orphan refusal caps at AUC 0.57–0.61 turns out to be a statement about *coarse units*: at constant deletion size, detection rises monotonically with k (`centroid_sbert` $0.564 \to 0.795 \to 0.984$ for $k = 10, 50, 200$). Strip the name and the ladder flattens to $0.560 \to 0.572 \to 0.623$ — $+0.063$ across a $20\times$ granularity change, against $+0.367$ on gold-form queries (App. F).

So does everything else that made the pattern look safe (Table 5 collects the cells). Routing accuracy collapses $0.966 \to 0.343$, and `key_exact` — a name matcher and nothing else — routes at 0.025 on every name-free surface. Deletion locality is lexical too: RDR is 0.000 on gold-form queries at $k = 200$ (against 5.8% published at $k = 10$, so fine units genuinely do help locality) but 0.092 name-stripped — deleting 20 of 200 sources moves 9.2% of *retained* queries to a different expert, worse than the coarse-unit number it was supposed to improve on.

One exception, and one attack. `ppl` does not collapse (0.783 name-stripped). The two behavioural selectors that do collapse read activation *magnitudes*; the survivor reads the expert's *loss*. The leak is therefore a property of the cheap selectors people deploy rather than something unavoidable, which is what makes a record-free defense worth pricing (§6). Meanwhile, because routing is lexical it is *attackable*: appending one chosen author name to a query captures 97.7% of queries against `key_exact` and 31.7% against `key_tfidf`, and *substituting* the name captures $\sim 87\%$ in every family — so an adversary chooses *whose* biography is attributed to the erased person.

6 The defense does not deploy

Does the one surviving detector yield a record-free refusal gate? The defender must be record-free, because the question is whether the system can refuse an orphan *without* maintaining exactly the registry of deleted people that deletion was supposed to eliminate. Scoring `ppl` over all 180 survivors costs 180 forward passes per query, and a free `key_tfidf` prefilter to the top-m candidates removes that cost with nothing measurable given up — $45\times$ at $m = 4$, $90\times$ at $m = 2$ — while *improving* detection, because the score is a `min` over candidates and each extra expert is another chance for an orphan to find an accidental good fit (App. G).

And it does not matter. At a 0.90 orphan catch rate the best achievable false-refusal rate on retained traffic is 0.000 on `gold` queries and 0.418 on `name_stripped` ones. A gold-form query *names the author being asked about*; the adversary who matters does not, and against them the gate refuses more than two fifths of everyone else's traffic. **Cost was never the binding constraint** — we removed it, 45–$90\times$, and the frontier did not move; it is bounded by discrimination. Nor is training longer a mitigation: at fixed rank, raising epochs $5 \to 25 \to 50$ drives `activation_norm` to 0.515 (chance) while `ppl` stays at 0.996, hiding the deletion signal from the weak probes and leaving it intact for the strong one (App. H).

7 Conclusion

Modular unlearning delivers what it promises: after deletion, no surviving parameter was trained on the removed source. It does not deliver what the word "deleted" is heard to mean. The queries survive the module, the selector reassigns them, and the system answers them with a stranger's life — specifically, fluently, and without refusing — while TOFU's forget metric scores that outcome at or above genuine deletion. A routed system's deletion claim is therefore incomplete without a *substitution report*: what fraction of orphaned queries are answered (ORR), with whose facts (CSAR), and at what collateral cost to retained traffic (RDR) — each on a query surface that does not name the deleted party. Deleting the module is the easy half. Deciding what the system says next is the deletion problem that modular architectures have quietly left open.

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A Full experimental setup

Pool. Llama-2-7B-chat as the frozen base, with $k = 200$ LoRA experts [Hu et al., 2022], one per TOFU author, rank 32, 25 epochs, lr 10^{-4}. TOFU's 200 fictitious authors with 20 QA pairs each give a clean ground truth for who knows what — which is what makes the fact-level CSAR metric computable at all. Deletion is TOFU `forget10`: authors 180–199, i.e. 20 of 200 sources and 400 orphaned questions. Two further pools (rank 8 / 5 epochs, rank 32 / 5 epochs) and a third built for App. H (rank 32 / 50 epochs) separate recipe effects from the pattern.

Selectors. Eight, in three families. *Lexical*: `key_exact` (author-name substring match), `key_tfidf`. *Feature-space*: `centroid_sbert` (MiniLM question embeddings [Reimers and Gurevych, 2019]), `centroid_lm` (base-model embeddings with adapters disabled). *Behavioural*: `ppl` (per-expert loss on the query), `activation_norm`, `attn_norm`, `logit_div`.

A property worth stating once: *no feature-space selector reads any expert's weights*, which we verified by finding their score matrices byte-identical across pools differing in rank and epochs. Recipe questions can therefore only be asked of the behavioural family — a control over a variable the measurement does not consume is not a control (App. I).

Evaluation protocol. All detection numbers use an author-parity split: even-numbered authors fit, odd-numbered authors evaluate, so every evaluated deleted source is one the detector never saw. Features are permutation-invariant functions of the sorted survivor scores, never raw column identities, so a detector fit on one deletion transfers to an unseen one. Every serving arm audits the *authors* taking each route before any metric is read — a plausible-but-wrong route is the failure mode these arms are most exposed to, and no downstream metric would flag it.

Threat model. The **adversary** has query access only: no deletion record, no list of which sources exist, no model weights, no training data. In the strongest variant we allow they also observe the selector's own scores over the *surviving* modules — which any system exposing retrieval confidence or expert attributions already hands out. Their goals are D3 and steering. The **defender** is likewise record-free, for the reason given in §6.

B Metric definitions in full

ORR — orphan response rate. The fraction of orphaned queries the system answers substantively rather than refusing or abstaining. Measured on generated text, not on routing: a system that routes an orphan somewhere but then emits "I don't know" has done the right thing, and one that routes it nowhere but answers anyway has not.

CSAR — cross-source attribution rate. The fraction of orphan answers that assert at least one fact that is (i) *distinctive* to the surviving source the query was routed to (appears in the gold answers of at most two of the 200 sources), (ii) *not* the deleted source's own fact — serving the deleted source's real content is a disclosure of the deleted source, a different and worse failure that must not be folded in — and (iii) *not* present in the frozen base model's answer to the same question, so the base's own knowledge is not credited to the selector.

We decompose CSAR into a *substantive* part (at least one matched fact is not merely a name-form of the routed survivor) and a *name-only* part. Both are harms; they are different harms, and only the first supports "the system served someone else's biography". Because 18 of 200 authors yield no extractable name (App. E), "substantive" is an upper and "name-only" a lower bound.

Why not n-grams. A ROUGE-based audit of the same orphan answers files 95.5% of them as novel confabulation with no leak. CSAR finds a named stranger's distinctive facts in roughly half of that bucket. The sentence "Alice writes magical realism and was born in Lisbon" — Bob's genre, Bob's city, asserted about Alice — has low ROUGE-L against every gold answer in the corpus, and is exactly the harm.

RDR — retained displacement rate. The fraction of *retained* queries that change module as a side effect of a deletion nobody asked for. Deletion is supposed to be local to the deleted source; RDR measures whether it is.

Detection AUC. Whether an orphan query can be told from a retained one using only the surviving selector's scores, with no deletion record consulted. This operationalises D3 and, read from the defender's side, is the ceiling on any record-free refusal gate.

Full CSAR decomposition. Table 3 gives the split quoted in §4.

Table 3: CSAR decomposition on `gold` queries, 400 orphans per router.

Router	CSAR	**substantive**	name-only
`centroid_sbert`	0.3325	**0.2400**	0.0925
`key_tfidf`	0.3650	**0.2950**	0.0700
random floor	0.2200	**0.1725**	0.0475

C Query transforms

A harm measured only on queries that name the deleted person is worth as much as a defense measured that way. Every result is reported across the axis in Table 4; the perturbation is applied to *both* classes, so no detector can learn "perturbed ⇒ deleted".

Table 4: The query-transform axis. `paraphrase` is included because it is the obvious robustness check and it reports a reassuring nothing: it preserves the author's name (coverage 0.900 vs. 0.895 for the gold form), so it perturbs everything except the feature the selectors actually use.

Transform	Construction	Role
`gold`	TOFU's question verbatim (names the author in $\sim$90% of rows)	anchor
`paraphrase`	TOFU's own paraphrase	**null by construction** — keeps the name
`name_stripped`	all name forms removed	**the hard case**
`indirect`	name replaced by a description built from the author's own facts	weaker than it looks
`para_stripped`	TOFU's paraphrase, name removed	the independent hard surface

The apparent hard case was not hard. `indirect` replaces the name with a description built from the author's *own* distinctive facts, which in practice prepends their book titles. It is therefore nearly as lexically self-identifying as the gold form: `key_tfidf`'s top-2 contains the true owner 0.965 of the time, against 0.998 gold and only 0.685 name-stripped. Decomposing routing accuracy on `key_tfidf`, with the name-keeping paraphrase as the control showing that rewording alone does not hurt ($0.973 \rightarrow 0.989$): removing the name costs -0.413, additionally rewording the rest costs a further -0.090, but `indirect` costs only -0.254 — the description hands $\sim$$+0.25$ back. *A transform built from the target's own facts cannot test anonymity.* We introduce `para_stripped` (TOFU's own paraphrase with names removed; median 12 words, $0/800$ retaining any name form) as the honest independent surface, and report `indirect` beside the gold-form column rather than as the hard case.

D The blind-metric construction in full

Construction. All 400 orphaned queries are rerouted to one fixed surviving expert; every expert, including the 20 supposedly forgotten ones, remains in the pool and in memory. Seven destination choices and one genuine-deletion baseline (experts dropped, orphans served by the frozen base) are scored at the same tier against the same held-fixed KS reference, so all eight cells are the same statistical test. Route audits confirm `deleted: 0 / rerouted: 1320` in every reroute arm against `deleted: 1320 / rerouted: 0` in the baseline, and `model_utility` is identical to 0.8009 in all eight.

The affinity hypothesis is refuted. Spearman ρ between `forget_quality` and destination affinity is $+0.109$ ($p = 0.82$), nominally opposite in sign to what the hypothesis predicts; it fails at a coarser evaluation tier too ($\rho = -0.059$, $p = 0.88$).

Paired intervals. The arms score identical rows (inter-arm $r = 0.88$–0.94), so this is a paired design. Resampling row indices once per iteration and applying that draw to every arm gives spread 95% CI $[0.2245,\ 0.6975]$, $P(\text{spread} > 0.25) = 0.961$, and at-or-above-deletion count $6/7$ with 95% CI $[2,\ 7]$.

Two limits. First, the *ordering* does not reproduce across evaluation tiers (Spearman extended$\sim$smoke $= +0.620$, $p = 0.14$): the spread replicates, the per-destination ranking does not, so no claim may name a winning destination. Second, a single `forget_quality` cell is worth about ± 0.35 (marginal 95% CI widths 0.63–0.72). The KS statistic moves on an exact lattice of $1/\mathrm{lcm}(n, m)$ — here $1/120$, one forget question — with roughly 30 attainable p-values above 0.05. The four decimals this literature reports are spurious precision.

E Where orphans go

Orphans are all reassigned. $400/400$ deleted-author questions are answered by some surviving expert under every selector. At $k = 200$ the mass is diffuse — busiest survivor 0.11–0.19, effective number of destinations 17.4–24.2 — except for `key_exact`, which sends 100% of them to one unit, because it returns the first shard *by index* whose name appears and falls back to `candidates[0]` otherwise.

Magnet saturation is refuted. A natural prediction is that deleting sources one at a time makes one survivor progressively the answerer for a growing share of the corpus. It does the opposite in six of seven cells: with one author deleted, all 20 of their questions go to that author's single nearest survivor (share 0.55–0.75); with twenty deleted, each has a *different* nearest survivor, so the pooled share falls and the effective count rises to 17–29. The prediction holds only where the router cannot tell the deleted sources apart — and there it holds sharply: `key_tfidf` on `indirect` queries saturates at 0.902 with an effective count of 1.2, one survivor answering nine of every ten orphans.

That survivor is a structural artifact. It is one of **18 of 200 authors from whom no name can be extracted**. Its centroid is the most generic point in the space; it already absorbs 68% of real-author and 45% of world-facts out-of-distribution queries. The routing magnet and the missing-name artifact are the *same* units, so any survivor-conditioned statistic is least trustworthy in exactly the name-free conditions this paper most wants to report. We flag every such cell rather than dropping it.

F Detectability, granularity, and steering

The granularity ladder. At constant deletion size, detection rises monotonically with k: `centroid_sbert` $0.564 \rightarrow 0.795 \rightarrow 0.984$ and `centroid_lm` $0.502 \rightarrow 0.628 \rightarrow 0.761$ for $k = 10, 50, 200$, with `key_tfidf` saturated throughout ($0.973 \rightarrow 0.989$). Attribution recall rises $0.300 \rightarrow 0.700 \rightarrow 1.000$.

Name-stripping flattens it. $0.560 \rightarrow 0.572 \rightarrow 0.623$: $+0.063$ across a $20\times$ granularity change, against $+0.367$ on gold-form queries. `key_tfidf` gains nothing at all (-0.021). At $k = 200$ detection falls $0.991 \rightarrow 0.623$, back inside the coarse-unit band that granularity was supposed to escape. The behavioural family replicates this: `activation_norm` $0.877 \rightarrow 0.495$ and `attn_norm` $0.758 \rightarrow 0.519$, i.e. chance, on both $k = 200$ pools. `ppl` does not: $0.783/0.769$ name-stripped on the rank-32 pools.

Table 5: What removing the author's name from the query does, at $k = 200$ — the summary behind §5. Detection AUC is orphan-vs-retained from the surviving selector's scores alone, with no deletion record consulted. Every property that makes the pattern look safe is a property of the query naming its source, except `ppl`, which reads the expert's *loss* rather than activation magnitudes.

Quantity	Selector	`gold`	`name_stripped`
detection AUC	`centroid_sbert`	0.991	0.623
	`key_tfidf`	0.989	0.968
	`activation_norm`	0.877	0.495
	`attn_norm`	0.758	0.519
	`ppl`	0.999	**0.783**
routing accuracy	`centroid_sbert`	0.966	0.343
	`key_exact`	0.998	0.025
RDR (lower better)	`centroid_sbert`	0.000	0.092

A learned probe adds nothing. Fitting a logistic reader over permutation-invariant functions of the surviving scores reaches AUC 0.990 at $k = 200$ with no deletion record — and its median lift over plain confidence is $+0.001$. D3 fails, but it fails to a threshold, not to a learned residual trace. Under every hard transform the same probe sits at its own shuffle control (0.446 vs. 0.503; 0.417 vs. 0.501; 0.414 vs. 0.532).

Steering the selector. Appending a single chosen author name to a query captures 97.7% of queries against `key_exact`, 31.7% against `key_tfidf`, but only 3.5% against `centroid_sbert`; *substituting* the name captures $\sim 87\%$ in every family.

G The defense frontier

1. It can be made cheap. Scoring `ppl` over all 180 survivors is 180 forward passes per query. A free `key_tfidf` prefilter cutting to the top-m candidates costs nothing measurable: $m = 4$ ($45\times$) under `name_stripped`, $m = 2$ ($90\times$) under `indirect`. Detection does not require *correct* routing — at $m = 4$ the prefilter loses the query's own expert 23% of the time and detection is unaffected, because orphans are uniformly poorly fit across whatever candidates are scored.

2. The prefilter improves detection rather than trading it away. Under `indirect`, AUC is monotone *decreasing* in m (0.9555 at $m = 2$ down to 0.8545 at $m = 180$). The score is a `min` over candidates, so each additional expert is another chance for an orphan to find an accidental good fit; restricting the candidate set denies orphans that lottery while retained queries keep their own expert. Cost and accuracy point the same way, which is rare enough to state plainly.

3. It is not deployable. An AUC cannot say whether a gate is usable. Table 6 gives the best achievable false-refusal rate on *retained* traffic, minimised over m. The gate is perfect where it is not needed and unusable where it is.

Table 6: False-refusal rate on retained traffic at a given orphan catch rate, for the `ppl` refusal gate.

Pool	Transform	catch 0.50	catch 0.90	catch 0.99
r32/e25	`gold`	0.000	**0.000**	0.000
	`indirect`	0.019	0.125	0.236
	`name_stripped`	0.120	**0.418**	0.654
r8/e5	`name_stripped`	0.361	0.639	0.769

A warning that must travel with the cost result. On the rank-8 pool the cheap gate "matches full" only because full is *poor* (0.63/0.65 AUC). A $90\times$ speedup at AUC 0.62 is not a defense.

H Training longer moves the leak; it does not remove it

Behavioural detectability is strongly recipe-dependent, and the axis is training *duration*, not capacity: holding epochs at 5 and raising rank $8 \rightarrow 32$ does not degrade `activation_norm` ($0.877 \rightarrow 0.934$), while holding rank at 32 and raising epochs $5 \rightarrow 25$ collapses it ($0.934 \rightarrow 0.608$). We built a third pool to add a fixed-in-advance third point.

Table 7: Gold-form detection AUC at fixed rank 32 across the epochs axis. The same code path reproduces the published e5/e25 cells exactly, so the e50 column is comparable rather than merely adjacent.

Selector	e5	e25	**e50**	reads
`ppl`	1.000	0.999	**0.996**	the expert's **loss**
`activation_norm`	0.934	0.608	**0.515**	**magnitudes**
`attn_norm`	0.700	0.554	**0.569**	**magnitudes**

`activation_norm` falls monotonically to 0.515 — **chance**. Read alone this is the cheapest defense in the literature, costing nothing but epochs. The same table refutes that reading: `ppl` is untouched at 0.996. Training duration does not remove the deletion signal; it removes it from the cheap magnitude-reading selectors and leaves it fully intact for the one that reads behaviour. **"Train longer" must not be offered as mitigation.** This also makes `ppl` the exception on two independent axes — query transform and training duration — which is why §6 rests on `ppl` specifically rather than on "behavioural selectors" as a class. One counter-current: name-stripped `ppl` declines gently along the same axis ($0.783 \rightarrow 0.769 \rightarrow 0.737$), so the exception is far more robust, not immune.

I Measurement constraints

Each constraint below invalidated a reading we had already written down. We state them because they generalise to anyone auditing a routed system.

1. **A paired quantity needs a paired interval.** The destination arms score identical rows (inter-arm $r = 0.88$–0.94). Bootstrapping each arm's marginal re-adds their *shared* question noise once per arm and declares any spread unresolvable. Our own first verdict on §3 made exactly this error, and it flipped on the paired test.
2. **Quote the exact resolution, not a sampled one.** The "achievable p-values" of a two-sample KS test, enumerated by random draw, grow with the draw count (73 $\rightarrow$ 88 between 2k and 60k draws, unconverged) — so such a count states nothing. Every attainable D is a multiple of $1/\mathrm{lcm}(n, m)$; use that.
3. **Structural holes in the corpus concentrate in the conditions you most want to report.** The 18 name-free authors distorted three separate results, and they are the same units that act as routing magnets. Flag or exclude them in any per-survivor statistic.
4. **A transform built from the target's own facts cannot test anonymity** (App. C).
5. **A sub-chance AUC means nothing without its own shuffle control.** Two apparent systematic sign flips were noise; the control spans 0.336–0.532 here, so differences below ~ 0.1 are not resolvable and must not be quoted as values.
6. **A control over a variable the measurement does not consume is not a control.** Our first recipe ablation was vacuous by construction: feature-space selectors read no expert weights, so three pools returned byte-identical matrices.

Defects that produced plausible numbers. Eight, all recorded in the campaign log rather than quietly fixed, because every one of them yielded output that looked right: a lazy adapter cache that silently scored non-resident experts as exactly 0.0; a route audit whose assertion ran *before* the artifact was written, destroying the arm it audited; a cache race that let a sibling arm read a 0-byte file; a consolidator that paired results with the wrong generation dump; an entire query condition made unreproducible by `sorted(set, key=len)` under hash randomisation; and the two statistical errors above. The countermeasure that kept working was computing the same quantity a second way.

J Extended related work

Exact unlearning by construction. SISA [Bourtoule et al., 2021] converts deletion into bounded retraining by sharding; LegoNet [Yu et al., 2022], S3T [Basu Roy Chowdhury et al., 2025], FedSGT [Zhang et al., 2025] and continual-learning forgetting services [Huang et al., 2025] carry the recipe — isolation plus bounded recomputation — into adapters, PEFT slices, federated rounds and analytic classifiers. Merging-based variants [Ilharco et al., 2023, Yu et al., 2024, Kuo et al., 2025] make deletion an exact subtraction, and a parallel line moves the adapted knowledge out of the weights entirely, into context that can be edited [Pawelczyk et al., 2024] or removed exactly [Muresanu et al., 2024]. All of these guarantee D1. None of them, to our knowledge, reports what the composition does with the queries that used to be served by the removed component — and the in-context variants inherit the same open question, since a retriever is a selector.

Routing and serving many experts. RAMoLE [Zhao et al., 2024] retrieves and composes uploaded LoRA experts; Compress-then-Serve [Brüel Gabrielsson et al., 2025] makes thousands of adapters servable; SEA [Schneider et al., 2026] makes deletion an artifact drop; memory adapters [Grimes et al., 2026] make it a block-list entry; SEUF [Zhuang et al., 2024] asks whether editing one MoE expert suffices, and finds the gate itself is the obstacle. Our results say the selector is not only an obstacle to unlearning *inside* a module but the component that determines what a deleted source's queries are answered *with*.

Localization. Maini et al. [2023] show memorization is localizable but not layer-confined; gradient routing [Cloud et al., 2024], MemSinks [Ghosal et al., 2025] and capability-removal-by-localization [Shilov et al., 2025] engineer localization so that a component can be removed. This paper is orthogonal and complementary: it grants perfect localization and asks what the surrounding system does afterwards.

Benchmarks and attacks. TOFU [Maini et al., 2024] is the yardstick, and §3 is a negative result about it in the routed setting specifically. Wu et al. [2025] show that exact unlearning of the final model does not prevent extraction from the broader pipeline; our failure is complementary and requires no pipeline access at all — the deployed system emits it in the normal course of answering.

K Limitations and what we do not claim

- **One benchmark, one base model, one deletion size.** TOFU's fictitious authors give clean ground truth about who knows what, which is exactly why the fact-level CSAR metric is computable at all; they are also uniform, synthetic and English. We deliberately do not claim these rates transfer to a heterogeneous production corpus.
- **We do not claim any destination is better or worse than another.** The per-destination ordering does not reproduce across evaluation tiers, only the spread does.
- **CSAR is not yet validated**: $\sim$300 hand labels are outstanding. The decomposition into substantive and name-only inherits the classifier's errors, and because 18 authors yield no extractable name, "substantive" is an upper and "name-only" a lower bound.
- **We do not claim no record-free defense is possible.** App. G bounds `ppl`-as-refusal-gate at $k = 200$ on three pools under one transform family.
- **We do not claim training duration mitigates anything** — App. H is a warning against precisely that reading.
- **The privacy/MIA column is withheld.** In one wave every arm reported byte-identical attack AUCs while serving different models, which we attribute to all queries falling to the out-of-distribution path; rather than publish a number we cannot explain, we report its absence.
- **The behavioural family has not been run under `para_stripped`**, the honest independent hard surface (App. C); §6's frontier is therefore stated on `name_stripped`.

Reproducibility. Every number in this paper is produced by a CPU-gated analysis over score matrices and generations that are committed alongside the code, with a per-claim map from each number to the artifact that produced it. The anonymized artifact URL will be inserted at submission time.